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VA-RECOVER: Extraction of Sub-Pixel Displacement for Inexpensive Visual Audio Recovery in Mobile Handset Cameras

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Abstract

The collision between sound pressure waves and physical surface boundaries produces micrometric oscillations that encode acoustic frequency spectra onto nearby objects. Conventional visual microphones measure these physical displacements using expensive industrial cameras and complex steerable pyramids. However, traditional 8-orientation steerable pyramids are ~21x overcomplete and incur massive performance overhead. By leveraging the CMOS rolling shutter mechanism, we can achieve an effective sampling frequency of several kilohertz from regular videos (usually 30-60 FPS). Combining CMOS rolling shutter capture with the 2D Dual-Tree Complex Wavelet Transform (DTCWT), which is only ~4x overcomplete, yields a ~5x computational speedup while maintaining the approximate shift-invariance necessary to estimate the sub-pixel displacement. Phase differences are extracted via complex conjugate multiplication, weighted by squared amplitude to suppress noise from untextured spatial regions before zero-phase temporal filtering. Measuring the signal-to-noise ratio (SNR) and speech intelligibility of reconstructed audio waveforms across changing sound pressure levels demonstrates that accessible consumer smartphone sensors can perform passive optical audio recovery.

Introduction and Research Problem

Non-contact optical sensing enables remote acoustic and vibration measurements without contact hardware. Sound pressure waves colliding with surface boundaries cause microscopic vibration motion that maps the frequency range of the sound source. Compared to standard contact hardware, such as accelerometers, optical vibrometry requires only a direct line of sight, eliminating mass-loading distortions on lightweight targets. Nonetheless, early visual audio recovery systems relied on high-speed cameras (2,000+ FPS) and intricate steerable pyramids (~21x overcomplete), causing substantial, unwanted memory and computational overhead [5], [13]. Contemporary mobile handset cameras employ CMOS rolling shutter sensors that read out pixels line-by-line, converting a regular video (30-60 FPS) into an efficient high-frequency temporal sampling engine at several kilohertz [1]. Integrating a lightweight 2D Dual-Tree Complex Wavelet Transform (DTCWT) phase extraction framework directly on the video taken with the rolling shutter narrows the divide between expensive and complex setups and enables convenient real-world visual audio recovery.

Research Questions and Hypotheses

We will assess sub-pixel phase sensitivity across spatial transform frameworks to determine the achievable reconstruction accuracy of mobile optical audio recovery.

> **Central Research Question**: To what extent can local phase extraction using a 2D Dual-Tree Complex Wavelet Transform (DTCWT) on consumer smartphone rolling shutter video recover intelligible acoustic signals compared to traditional complex steerable pyramids?

The **independent variable (IV)** is the spatial decomposition framework (2D DTCWT vs. Complex Steerable Pyramid), while the **dependent variable (DV)** is the recovered audio Signal-to-Noise Ratio (SNR in dB).

- The **null hypothesis (H_0)** states that there is no statistically significant difference in mean recovered audio SNR between 2D DTCWT extraction on rolling shutter video and complex steerable pyramid extraction. ($H_0: \mu_{DTCWT} - \mu_{Steerable} = 0$)
- The **alternative hypothesis (H_1)** predicts that 2D DTCWT extraction will achieve statistically non-inferior audio SNR ($\geq -3\ dB$) with significantly lower computational latency compared to complex steerable pyramids. ($H_1: \mu_{DTCWT} - \mu_{Steerable} \geq -3\ dB$)

Rigorous and repeated hypothesis testing establishes whether lightweight wavelet transforms preserve sufficient local phase coherence for acoustic reconstruction on resource-constrained mobile platforms.

Literature Review

Modern non-contact optical vibrometry relies on phase-based motion processing, a foundational framework established by Wadhwa et al., who showed that sub-pixel physical motion equates directly to linear phase variations across complex spatial subbands [11]. Building upon the groundbreaking visual microphone established and developed by Davis et al., which proved that acoustic pressure waves induce micrometric mechanical surface vibrations on everyday objects, these displacements can be recovered from video via multi-scale complex steerable pyramids [13]. To avoid the requirement of expensive high-speed industrial cameras, subsequent work by Davis et al. demonstrated that the CMOS rolling shutter sequential sweeping converts a regular video into a high-frequency temporal sampling engine operating at several kilohertz [1].

Regardless of these breakthroughs, early implementations faced severe memory consumption and slow computational latencies on standard processors. To mitigate this computational load, Wadhwa et al. introduced compact Riesz pyramids for real-time phase-based motion processing [12]. However, Riesz pyramids' directional selectivity is less robust compared to multi-orientation complex filter banks.

Researchers experimented with different hardware and algorithmic framework alternatives, such as dual-shutter systems or neuromorphic event cameras [3], [4]. To tackle spatial phase stability, recent open-source implementations by J. Jose [6] utilize the 2D Dual-Tree Complex Wavelet Transform (DTCWT) to measure shift-invariant spatial phase features. Leveraging 2D DTCWT allows for a ~5x computational speedup compared to steerable pyramids while preserving high phase sensitivity for sub-pixel displacement estimation.

Even with the computational efficiency of 2D DTCWT for motion analysis in high-speed cameras, existing literature lacks a full paradigm that clearly combines the CMOS rolling shutter mechanism with 2D DTCWT spatial phase extraction. In addition to that, no benchmark data

16 have assessed the signal-to-noise ratio (SNR) and speech intelligibility of audio recovery from the visual output of everyday consumer mobile devices using this efficient wavelet pipeline.

Proposed Methodology: The method is a quantitative approach.

We test everyday objects such as paper boxes and chip bags to measure and determine how sound pressure levels drive sub-pixel phase shifts. Each target is placed under controlled conditions, sits in a dead room, and experiences calibrated frequency sweeps from 100 Hz to 1 kHz at a constant 75 dB SPL.

Hardware & Video Specifications

- **Camera Sensor Platform**: Commercial smartphone camera sensor recording at 1080p (or better) and 30/60 FPS.
- **Rolling Shutter Readout**: The CMOS image sensor operates with a line-by-line rolling shutter readout time of $t_{row} \approx 15 - 30\mu s\ per\ row$.
- **Effective High-Rate Sampling**: Row-sequential acquisition yields an effective temporal sampling frequency of $f_{line} = 1/t_{row} \approx\ 33kHz$ to $66kHz$ (or per-line sampling rate $\geq\ 15kHz$) allowing the system to capture high-frequency acoustic surface vibrations well above standard video Nyquist limits without specialized high-speed cameras.
- **Platform Stabilization**: The camera platform is rigidly mounted on a heavy-duty tripod to eliminate low-frequency platform egomotion.

VA-RECOVER Signal Processing Architecture

Step	Module	Mathematical / Architectural Description
Step 1	Dual-Tree Wavelet Filter Bank	Decomposes each frame using 2D DTCWT. Level 1 utilizes Near-Symmetric 13/19-tap biorthogonal wavelet filters (*near_sym_b*), while Levels ≥ 2 utilize 14/14-tap Quarter-Shift (Q-shift) filters (*qshift_b*). Produces 6 oriented complex sub-bands per scale level($\pm 15°$, $\pm 45°$, $\pm 75°$) with ~4x overcomplete [6].
Step 2	CMOS Rolling Shutter Line-Sampling	Exploits sequential row readout ($t_{row} \approx 15 - 30\mu s\ per\ row$) across the CMOS array. Converts row-wise pixel time-series into a high-rate temporal sequence (effective $f_{sample} \geq\ 15kHz$).
Step 3	Conjugate Local Phase Extraction	Computes local phase differences relative to the frame t_0 via conjugate multiplication: $C_t \cdot \bar{C}_{\text{ref}} = \lvert C_t \rvert \lvert \bar{C}_{\text{ref}} \rvert e^{i\Delta\phi_t}$. Eliminates static surface texture patterns and extracts pure sub-pixel displacement [11], [13].

Step 4	Amplitude-Squared (A^2) Weighting	Aggregates phase differences across spatial dimensions weighted by squared magnitude $A^2 = \|C_t\|^2$: $\Phi(s,\theta,t) = \sum A^2 \Delta\phi$. Suppresses noisy, untextured regions while highlighting reliable texture edges [11], [13].
Step 5	Zero-Phase Bandpass Filtering	Applies a 4th-order zero-phase Butterworth filter (*sosfiltfilt*) to the phase time-series. Suppresses camera drift (<100 Hz) and sensor thermal noise (>1 kHz) without temporal phase distortion.
Step 6	Cross-Correlation & Audio Export	Aligns phase signals across scales/orientations using cross-correlation lag $\tau(s,\theta)$. Sums aligned sub-bands into a single 1D waveform $\hat{s}(t)$ and exports a 16-bit WAV file sampled at f_{line}.

Data Analysis

Reconstructed audio waveforms and execution speeds will be evaluated using Signal-to-Noise Ratio (SNR in dB) and processing latency (ms/frame) across varying Sound Pressure Levels (65–85 dB SPL). Distributional normality will be evaluated using the Shapiro-Wilk test ($\alpha = 0.05$), guiding the choice between an independent-samples t-test (if normal) or a Mann-Whitney U test (if non-normal) to compare the 2D DTCWT pipeline against steerable pyramids. Statistical significance ($\alpha = 0.05$), and 95% confidence intervals will be used to assess non-inferiority bounds (≥ -3dB SNR) and computational performance equivalence.

Expected Outcome

The study is expected to demonstrate the effectiveness of the VA-RECOVER system as an inexpensive, lightweight visual audio recovery pipeline for the purpose of extracting acoustic vibrations from video footage recorded via a standard consumer smartphone. It will evaluate the overall computational efficiency offered by the 2D Dual-Tree Complex Wavelet Transform and the accuracy of the reconstructed audio signals across calibrated Sound Pressure Levels (65–85 dB SPL), as well as the noise suppression capabilities provided by A² spatial weighting alongside zero-phase Butterworth filtering. The resulting data will help guide future potential improvements in accessible, non-contact optical vibrometry for real-world acoustic sensing applications on standard mobile hardware.

Technical Limitations

Due to surface texture dependencies on smooth or glossy target objects, platform egomotion from camera shaking, and ambient interference from 50/60 Hz AC mains lighting flicker, the system may experience reduced phase signal-to-noise ratios or harmonic frequency distortion, especially in low-light environments or on untextured targets.

REFERENCES

[1] A. Davis, K. L. Bouman, J. G. Chen, M. Rubinstein, F. Durand, and W. T. Freeman, "Visual vibrometry: Estimating material properties from small motion in video," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 5335–5343, doi: 10.1109/cvpr.2015.7299171.

[2] Y. Zhang, S. L. Pintea, and J. C. van Gemert, "Video acceleration magnification," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 529–537, doi: 10.1109/cvpr.2017.63.

[3] M. Sheinin, D. Chan, M. O'Toole, and S. G. Narasimhan, "Dual-shutter optical vibration sensing," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 16362–16365, doi: 10.1109/cvpr52688.2022.01589.

[4] R. Niwa, T. Fushimi, K. Yamamoto, and Y. Ochiai, "Live demonstration: Event-based visual microphone," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2023, pp. 4054–4055, doi: 10.1109/cvprw59228.2023.00424.

[5] H.-Y. Wu, M. Rubinstein, E. Shih, J. Guttag, F. Durand, and W. T. Freeman, "Eulerian video magnification for revealing subtle changes in the world," *ACM Transactions on Graphics*, vol. 31, no. 4, pp. 65:1–65:8, Jul. 2012, doi: 10.1145/2185520.2185561.

[6] J. Jose, "Visual-Mic: 2D dual-tree complex wavelet transform for sub-pixel motion extraction," GitHub Repository, 2021. [Online]. Available: https://github.com/joeljose/Visual-Mic

[7] A. Davis, J. G. Chen, and F. Durand, "Image-space modal bases for plausible manipulation of objects in video," *ACM Transactions on Graphics (SIGGRAPH Asia)*, vol. 34, no. 6, pp. 1–7, Jan. 2015, doi: 10.1145/2816795.2818095.

[8] T.-H. Oh, R. Jaroensri, C. Kim, M. Elgharib, F. Durand, W. T. Freeman, and W. Matusik, "Learning-based video motion magnification," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 633–648, doi: 10.1007/978-3-030-01231-1_39.

[9] N. Wadhwa, "Revealing and analyzing imperceptible deviations in images and videos," Ph.D. dissertation, Dept. Elect. Eng. Comput. Sci., Massachusetts Institute of Technology, Cambridge, MA, USA, 2016.

[10] Z. Wang, X. Ji, J.-B. Huang, S. Satoh, X. Zhou, and Y. Zheng, "Neural global shutter: Learn to restore video from a rolling shutter camera with global reset feature," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 5811–5820, doi: 10.1109/cvpr52688.2022.00572.

[11] N. Wadhwa, M. Rubinstein, F. Durand, and W. T. Freeman, "Phase-based video motion processing," *ACM Transactions on Graphics*, vol. 32, no. 4, pp. 80:1–80:10, Jul. 2013, doi: 10.1145/2461912.2461966.

[12] N. Wadhwa, M. Rubinstein, F. Durand, and W. T. Freeman, "Riesz pyramids for fast phase-based video magnification," in *IEEE International Conference on Computational Photography (ICCP)*, 2014, pp. 1–10, doi: 10.1109/iccphot.2014.6831820.

[13] A. Davis, M. Rubinstein, N. Wadhwa, G. J. Mysore, F. Durand, and W. T. Freeman, "The visual microphone: Passive recovery of sound from video," *ACM Transactions on Graphics*, vol. 33, no. 4, pp. 79:1–79:10, Jul. 2014, doi: 10.1145/2601097.2601112.

[14] J. G. Chen and E. C. Stewart, "Video-based methods for measurement of vibration mode shapes," NASA Technical Report, NASA Marshall Space Flight Center & MIT Lincoln Laboratory, NASA/TM-20240012703, 2024.